



Vocational Education — Cbse Chapter Test — 20 Marks

ANSWER KEY — FOR GRADING

Subject: Vocational Education | Class: Class 8

Total Marks: 20 | Time Allowed: 0h 40m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	When uploading a programme to an Arduino Uno, what must be selected in the Tools menu under Port? (A) USB Port 1 (B) Serial Monitor (C) Arduino Uno Com (D) Board Manager Answer: Correct Option: (C) ► Why? C. After connecting the Arduino board to the computer, open the Tools menu, go to the port, and select Arduino Uno Com. The com number is selected automatically by the computer. [Ch 4: Project 4: Home Automation]	1
2	What are the three key steps in the automation cycle? (A) Switching, Connecting, and Breaking (B) Input (Sensing), Processing (Decision making), and Output (Action) (C) Designing, Drawing, and Printing (D) Building, Testing, and Selling Answer: Correct Option: (B) ► Why? B. The automation cycle consists of three key steps: Input (Sensing) where sensors detect changes, Processing (Decision making) where a microcontroller analyses input, and Output (Action) where the system carries out the action. [Ch 4: Project 4: Home Automation]	1



Q.NO.	QUESTIONS	MARKS
3	In Activity 2, students are instructed to make simulated circuits. Which components and their specifications are correctly mentioned in the chapter for this activity? (A) LED, 100 ohms resistor, DC battery 6V, and switch (B) Buzzer, 330 ohms resistor, AC battery 12V, and button (C) LED, 220 ohms resistor, DC battery 9V, switch and buzzer with same specifications (D) Motor, 470 ohms resistor, DC battery 5V, and relay Answer: Correct Option: (C) ► Why? The chapter specifically lists two circuit configurations: 'LED, 220 ohms resistor, DC battery 9V, switch' and 'Buzzer, 220 ohms resistor, DC battery 9V, switch'. Both use the same resistor value (220 ohms) and battery (9V DC). ► Step-by-step: 1. Look at Activity 2 under 'What do I have to do?' section 2. Find the exact component specifications listed 3. Note that both LED and buzzer circuits use 220 ohms resistor and 9V DC battery 4. Match these specifications with the given options [Ch 4: Project 4: Home Automation]	1
4	In an automatic garden irrigation system, what happens when the soil becomes too dry? (A) The motion sensor detects movement in the garden (B) The system turns off the water pump immediately (C) The moisture sensor sends a signal to turn on the water pump (D) The temperature sensor measures the soil temperature Answer: Correct Option: (C) ► Why? C. When the soil becomes too dry, the moisture sensor installed in the soil sends a signal to a water pump. The pump then automatically turns on and waters the plants. [Ch 4: Project 4: Home Automation]	1
5	Which example from nature demonstrates automation similar to GPS used by humans? (A) Lungs breathing in oxygen (B) Water evaporating from lakes (C) Stomach digesting food (D) Birds migrating without GPS Answer: Correct Option: (D) ► Why? D. Birds migrate without Global Positioning System (GPS), generally used by humans. This demonstrates animal instincts as a form of natural automation. [Ch 4: Project 4: Home Automation]	1



Q.NO.	QUESTIONS	MARKS
6	After writing a programme in Arduino IDE, Priya wants to upload it to her Arduino Uno board. What steps should she follow according to the chapter? (A) Connect the LED to analog pins and press the reset button (B) Open the tools menu, go to examples, and select basics (C) Change the value in the delay command and save the file (D) Connect the Arduino board to the computer using its cable, go to the tools menu, select the port, and click the upload arrow Answer: Correct Option: (D) ► Why? The chapter provides clear steps: connect the Arduino board to the computer using its cable, then open the tools menu, go to port and select Arduino Uno, and finally click on the arrow under the file menu to upload the programme. ► Step-by-step: 1. Step 1: Connect the Arduino board to the computer using its cable. 2. Step 2: Open the tools menu on the computer and go to the port. 3. Step 3: Select Arduino Uno and click on the arrow under the file menu to upload the programme. [Ch 4: Project 4: Home Automation]	1
7	Which of the following is an example of natural automation in the human body? (A) Reading a book (B) Heart beating to pump blood (C) Walking and running (D) Speaking to a friend Answer: Correct Option: (B) ► Why? B. The heart beats automatically to pump blood without conscious effort, just like an automated machine. This is an example of natural automation. [Ch 4: Project 4: Home Automation]	1
8	Which of the following is mentioned as a job related to automation work? (A) Chef (B) Automation engineer (C) Doctor (D) Mechanic Answer: Correct Option: (B) ► Why? B. Examples of jobs related to automation work mentioned are automation engineer, technical support specialist, and programmer. [Ch 4: Project 4: Home Automation]	1
Section B Attempt all 3 questions. Each carries 2 mark(s).		
9	Name three jobs related to automation work.	2



Q.NO.	QUESTIONS	MARKS
	Answer: Jobs related to automation work include automation engineer, technical support specialist, and programmer. These professionals work with automated systems and programming. ► Why? This tests awareness of career opportunities in the field of automation as mentioned in the source. [Ch 4: Project 4: Home Automation]	
10	What is the function of a Digital Multimeter and who commonly uses it? Answer: A Digital Multimeter is an electronic measuring tool used to check voltage, current, and resistance in electrical circuits. It is commonly used by engineers, electricians, and hobbyists to test and troubleshoot circuits. ► Why? The source provides both the measuring function and the typical users of this tool. [Ch 4: Project 4: Home Automation]	2
11	Explain the function of a breadboard in circuit prototyping. Answer: A breadboard helps connect different components without soldering them, allowing quick testing of circuits. If something does not work, parts can be rearranged easily. It is used only for prototyping before making the circuit permanent. ► Why? The source emphasizes the temporary, non-soldered nature of breadboard connections for testing purposes. [Ch 4: Project 4: Home Automation]	2
Section C Attempt 2 out of 3.		
12	What is the role of a programme in an automation system and why is it necessary? Answer: A programme is a set of instructions given to the microcontroller in an automation system. The microcontroller is the brain of the circuit that processes sensor information and instructs output devices to act. However, the microcontroller cannot function without proper guidance. Users need to give instructions to get the desired response from microcontrollers. This set of instructions is called a programme. Without a programme, the microcontroller would not know how to respond to sensor inputs or what actions to trigger. The programme tells the microcontroller exactly what to do when it receives specific inputs, making the automation system function intelligently and perform tasks automatically.	3



Q.NO.	QUESTIONS	MARKS
	► Why? The answer explains that a programme is a set of instructions for microcontrollers, necessary for them to process inputs and produce desired outputs. ► Step-by-step: 1. Step 1: Define programme as a set of instructions given to the microcontroller. 2. Step 2: Explain that microcontrollers need instructions to process sensor information. 3. Step 3: Describe how programmes tell microcontrollers what actions to take for specific inputs. [Ch 4: Project 4: Home Automation]	
	OR What steps are followed to program an Arduino Uno board to blink an LED? Answer: To program an Arduino Uno to blink an LED, first download the Arduino IDE programme from the official Arduino website. Start the Arduino IDE on the computer and go to File ⇒ Examples ⇒ Basics ⇒ Blink to open the Blink programme. Connect the Arduino board to the computer using its cable. Open the tools menu, go to port and select Arduino Uno Com. Click on the arrow under the file menu to upload the programme to the Uno board. This will blink the onboard LED every second. To change the blinking speed, modify the value in the delay command and upload the programme again. ► Why? The answer follows the sequential steps outlined in the source material for uploading the Blink programme to Arduino. ► Step-by-step: 1. Step 1: Explain downloading Arduino IDE and opening Blink programme. 2. Step 2: Describe connecting the board and selecting the correct port. 3. Step 3: Mention uploading the programme and modifying delay for different timing. [Ch 4: Project 4: Home Automation]	
13	Explain the working of a motion detection sensor and state one practical application. Answer: A motion detection sensor senses infrared radiation (heat) from objects like humans and animals. When someone moves in front of it, the sensor detects the heat change and sends a signal to trigger an action. For example, in a classroom with an automated power system, the motion sensor detects movement and keeps lights on. When no motion is detected, it signals the system to turn off the lights automatically. This helps save energy and reduce costs by ensuring lights are only on when someone is present in the room. The sensor works independently without manual operation. ► Why? The answer describes how motion sensors detect infrared radiation and trigger actions, with a classroom lighting example from the source. ► Step-by-step: 1. Step 1: Explain that motion sensors sense infrared radiation (heat) from objects. 2. Step 2: Describe how the sensor detects heat changes and sends signals. 3. Step 3: Give the practical example of automatic classroom lighting.	3



Q.NO.	QUESTIONS	MARKS
	[Ch 4: Project 4: Home Automation]	

